

Predicting Cell A's State-of-Health

Solved Numerical Problem on Real Data

Battery Degradation MTP — Kirkaldy Experiment 5 (LG INR21700-M50T)

Using the real measured and simulated SoH values for Cells D (25°C) and F (40°C), we carry out the entire prediction of Cell A (10°C): gap, inputs, standardisation, kernel, covariance matrix, weights, predicted mean and variance, the $\pm 2\sigma$ band, the log marginal likelihood, and the final RMSE. The measured data come from the Kirkaldy Experiment 5 cycle-ageing dataset; the simulated SoH values come from a PyBaMM DFN model run with the Optimized degradation parameter set. We model the simulation's error (the gap) with a Gaussian Process: this follows the established idea of correcting a computer model with a statistical discrepancy term [2], applied here in the spirit of GP regression for battery state-of-health [4]. Boxed text gives short plain-language notes or the graduate-level reasoning; results sit in their own boxes.

Why a 3-point subset? A Gaussian Process must invert an $N \times N$ matrix. To keep that matrix small enough to print and invert by hand, we use 3 representative cycles per training cell ($N = 6$). The full pipeline uses all 16 RPT points; the equations are identical.

1 Notation — every symbol used

Symbol	Meaning
n	ageing cycle number (x-axis)
T	cell temperature in kelvin (K); $1/T$ is the Arrhenius coordinate
$\text{SoH}_{\text{exp}}, \text{SoH}_{\text{sim}}$	measured and simulated State-of-Health (1.0 = fresh)
$g = \text{SoH}_{\text{exp}} - \text{SoH}_{\text{sim}}$	the gap (how wrong the simulation is)
$x = (n, 1/T)$	one model input point; x_s = standardised version
μ, σ	per-feature mean and standard deviation (for standardising)
$\bar{g}$	mean of the training gaps (used to centre the target)
$y = g - \bar{g}$	centred target the GP actually fits
σ_f	kernel signal scale (size of the wiggles); σ_f^2 its variance
ℓ_1, ℓ_2	length-scales for cycle and for $1/T$ (ARD: one each)
σ_n^2	noise variance (measurement scatter / regulariser)
$k(x, x')$	kernel: similarity between two points
K	$N \times N$ covariance matrix; $\tilde{K} = K + \sigma_n^2 I$
$\alpha = \tilde{K}^{-1} y$	weight vector
k_*	similarities of a new point to all training points
$m(x_*), v(x_*)$	predicted mean gap and its variance; $\sigma_* = \sqrt{v}$

2 Step 0 — The real data

These are the extracted values (one row per RPT). The simulation is the PyBaMM output, and the gap is their difference. The experimental columns are the measured Kirkaldy data; the simulated columns are the Optimized physics-model output.

	Cell D (25°C)			Cell F (40°C)			Cell A (10°C)		
n	SoH _{exp}	SoH _{sim}	gap	SoH _{exp}	SoH _{sim}	gap	SoH _{sim}	SoH _{exp}	gap
0	1.00000	1.00000	0.000	1.00000	1.00000	0.000	1.00000	1.00000	0.000
78	0.95806	0.98267	−0.025	0.95606	0.97159	−0.016	0.98159	0.95058	−0.031
156	0.93325	0.97103	−0.038	0.93442	0.95341	−0.019	0.96647	0.91861	−0.048
234	0.90831	0.95956	−0.051	0.90962	0.93692	−0.027	0.95135	0.88676	−0.065
312	0.88949	0.94799	−0.059	0.89643	0.92124	−0.025	0.93621	0.86394	−0.072
390	0.88346	0.93625	−0.053	0.88399	0.90598	−0.022	0.92105	0.84690	−0.074
468	0.87622	0.92427	−0.048	0.87329	0.89093	−0.018	0.90584	0.83210	−0.074
546	0.86895	0.91200	−0.043	0.86253	0.87602	−0.013	0.89056	0.82263	−0.068
624	0.86259	0.89937	−0.037	0.85223	0.86113	−0.009	0.87519	0.81435	−0.061
702	0.85528	0.88628	−0.031	0.84230	0.84616	−0.004	0.85972	0.80657	−0.053
780	0.84936	0.87263	−0.023	0.83221	0.83128	0.001	0.84412	0.80079	−0.043
858	0.84072	0.85820	−0.017	0.82101	0.81628	0.005	0.82842	0.79345	−0.035
936	0.83300	0.84275	−0.010	0.80965	0.80110	0.009	0.81263	0.78665	−0.026
1014	0.82560	0.82604	−0.000	0.79852	0.78567	0.013	0.79666	0.78256	−0.014
1092	0.81731	0.80799	0.009	0.78666	0.76991	0.017	0.78049	0.77576	−0.005
1170	0.81032	0.78859	0.022	0.76787	0.75374	0.014	0.76410	0.77252	0.008

The bold rows ($n = 234, 546, 858$) are the three cycles we use for the hand calculation. Cell A's experiment is shown for reference but is used only at the very end to grade the prediction.

Notice the gaps are biggest in the middle of life and shrink (even flip sign) late. Cell A (coldest) has the largest gaps of all — its simulation is the most optimistic. That is the error we must predict and remove.

3 Step 1 — The gap (simulation error)

Using the three chosen cycles, $g = \text{SoH}_{\text{exp}} - \text{SoH}_{\text{sim}}$. This gap is the simulation's discrepancy from reality — exactly the quantity a statistical model is built to predict when correcting a computer code [2]:

$$g_D = \begin{pmatrix} -0.05125 \\ -0.04305 \\ -0.01748 \end{pmatrix}, \quad g_F = \begin{pmatrix} -0.02730 \\ -0.01349 \\ +0.00473 \end{pmatrix}.$$

(For reference, A's true gaps are $-0.06459, -0.06793, -0.03497$ — larger than D and F, exactly because A is colder.)

4 Step 2 — Build the inputs and the temperature coordinate

Each point is $x = (n, 1/T)$. The three temperatures give

$$\frac{1}{T_D} = \frac{1}{301.75} = 3.31400 \times 10^{-3}, \quad \frac{1}{T_F} = \frac{1}{314.75} = 3.17712 \times 10^{-3}, \quad \frac{1}{T_A} = \frac{1}{289.45} = 3.45483 \times 10^{-3}.$$

Six training inputs (D's three cycles, then F's three) and their gaps:

$$X = \begin{pmatrix} 234 & 3.31400 \times 10^{-3} \\ 546 & 3.31400 \times 10^{-3} \\ 858 & 3.31400 \times 10^{-3} \\ 234 & 3.17712 \times 10^{-3} \\ 546 & 3.17712 \times 10^{-3} \\ 858 & 3.17712 \times 10^{-3} \end{pmatrix}, \quad g = \begin{pmatrix} -0.05125 \\ -0.04305 \\ -0.01748 \\ -0.02730 \\ -0.01349 \\ +0.00473 \end{pmatrix}.$$

We use $1/T$ because thermally activated degradation obeys the Arrhenius law, $\text{rate} = A e^{-E_a/RT}$, so $\ln(\text{rate})$ is linear in $1/T$. The same Arrhenius temperature dependence drives the degradation reactions in the underlying physics model. Placing temperature in this coordinate makes the GP’s smoothness assumption physically meaningful.

5 Step 3 — Standardise the inputs

Features have very different magnitudes, so rescale each with $x_s = (x - \mu)/\sigma$. Computed over the six points:

$$\mu = (546.0, 3.24556 \times 10^{-3}), \quad \sigma = (254.747, 6.844 \times 10^{-5}).$$

Example for cycle 234: $x_s = (234 - 546)/254.747 = -1.2247$. For $1/T_D$: $(3.31400 - 3.24556) \times 10^{-3}/6.844 \times 10^{-5} = +1.000$. Full result:

$$X_s = \begin{pmatrix} -1.2247 & +1 \\ 0.0000 & +1 \\ +1.2247 & +1 \\ -1.2247 & -1 \\ 0.0000 & -1 \\ +1.2247 & -1 \end{pmatrix}.$$

6 Step 4 — Centre the target

$$\bar{g} = \frac{1}{6} \sum_i g_i = -0.02464, \quad y = g - \bar{g} = \begin{pmatrix} -0.02661 \\ -0.01841 \\ +0.00716 \\ -0.00266 \\ +0.01115 \\ +0.02937 \end{pmatrix}.$$

We fit the GP on y and add $\bar{g}$ back at the end.

7 Step 5 — The kernel (similarity function)

The covariance function is the Matérn class with $\nu = 5/2$, in the form given by Rasmussen and Williams [3, Eq. 4.17], with a separate length-scale per feature (ARD, 3, Sec. 5.1):

$$k(x, x') = \underbrace{\sigma_f^2}_{\text{signal var.}} \underbrace{\left(1 + \sqrt{5}r + \frac{5}{3}r^2\right) e^{-\sqrt{5}r}}_{\text{Matérn-5/2}} + \underbrace{\sigma_n^2 \delta_{xx'}}_{\text{noise}}, \quad r = \sqrt{\left(\frac{x_1 - x'_1}{\ell_1}\right)^2 + \left(\frac{x_2 - x'_2}{\ell_2}\right)^2}.$$

Hyperparameters used here (fixed to clean values for hand-work; the full script learns them by maximising Step 10):

$$\sigma_f = 0.06 \ (\sigma_f^2 = 0.0036), \quad \ell_1 = 2.0, \quad \ell_2 = 3.0, \quad \sigma_n^2 = 1 \times 10^{-4}.$$

σ_f^2 sets the prior variance of the gap. ℓ_1, ℓ_2 are ARD length-scales: a separate “reach” for the cycle axis and the $1/T$ axis, so the model can decide each feature’s relevance. $\nu = 5/2$ gives twice-differentiable sample paths — smooth yet physical. The small σ_n^2 models measurement scatter and, crucially with only two temperatures, regularises $\tilde{K}$ so it stays invertible (here the condition number is ≈ 51).

Worked kernel value 1 — same temperature, different cycle.

D@234 (−1.2247, +1) vs D@546 (0, +1):

$$r = \sqrt{\left(\frac{-1.2247-0}{2.0}\right)^2 + 0^2} = 0.6124,$$

$$k_{5/2} = \left(1 + \sqrt{5}(0.6124) + \frac{5}{3}(0.6124)^2\right) e^{-\sqrt{5}(0.6124)} = (1+1.3694+0.6250)e^{-1.3694} = 2.9944 \times 0.2543 = 0.7614,$$

$$k = 0.0036 \times 0.7614 = 0.002741.$$

Worked kernel value 2 — same cycle, different temperature.

D@234 (−1.2247, +1) vs F@234 (−1.2247, −1):

$$r = \sqrt{0 + \left(\frac{1-(-1)}{3.0}\right)^2} = \frac{2}{3} = 0.6667,$$

$$k_{5/2} = (1+1.4907+0.7407)e^{-1.4907} = 3.2314 \times 0.2252 = 0.7278, \quad k = 0.0036 \times 0.7278 = 0.002620.$$

8 Step 6 — The covariance matrix

Evaluating the kernel for all 6×6 pairs:

$$K = \begin{pmatrix} 0.00360 & 0.00274 & 0.00145 & 0.00262 & 0.00209 & 0.00117 \\ 0.00274 & 0.00360 & 0.00274 & 0.00209 & 0.00262 & 0.00209 \\ 0.00145 & 0.00274 & 0.00360 & 0.00117 & 0.00209 & 0.00262 \\ 0.00262 & 0.00209 & 0.00117 & 0.00360 & 0.00274 & 0.00145 \\ 0.00209 & 0.00262 & 0.00209 & 0.00274 & 0.00360 & 0.00274 \\ 0.00117 & 0.00209 & 0.00262 & 0.00145 & 0.00274 & 0.00360 \end{pmatrix},$$

then add noise on the diagonal: $\tilde{K} = K + \sigma_n^2 I$ (each 0.00360 $\rightarrow$ 0.00370).

9 Step 7 — Solve for the weights

$\alpha = \tilde{K}^{-1}y$, the weight vector of the GP predictor [3, Eq. 2.25 and Algorithm 2.1]. Inverting numerically and multiplying by y :

$$\alpha = (-6.626, -13.219, +3.364, +3.587, +3.111, +11.401)^\top.$$

In code one solves $\tilde{K}\alpha = y$ via the Cholesky factor rather than forming the inverse, for $O(N^3)$ stability. The weights can be large and of mixed sign; they encode how each training residual pushes neighbouring predictions.

10 Step 8 — Predict Cell A

A's standardised temperature: $(3.45483 - 3.24556) \times 10^{-3} / 6.844 \times 10^{-5} = +3.0577$. So A points are $(n_s, +3.0577)$

For A@234, $x_* = (-1.2247, +3.0577)$, the similarity vector is

$$k_* = (0.002577, 0.002057, 0.001158, 0.001237, 0.001040, 0.000647).$$

Predicted mean gap $m = k_*^\top \alpha + \bar{g}$ [3, Eq. 2.25]:

$$\begin{aligned} k_*^\top \alpha &= (0.002577)(-6.626) + (0.002057)(-13.219) + (0.001158)(3.364) \\ &\quad + (0.001237)(3.587) + (0.001040)(3.111) + (0.000647)(11.401) = -0.02532, \end{aligned}$$

$$m = -0.02532 + (-0.02464) = -0.04996.$$

Predicted variance $v = k(x_*, x_*) - k_*^\top \tilde{K}^{-1} k_*$ [3, Eq. 2.26]:

$$v = 0.0036 - 0.002019 = 1.581 \times 10^{-3}, \quad \sigma_* = \sqrt{v} = 0.0398.$$

($k(x_*, x_*) = \sigma_f^2 = 0.0036$ since the distance to itself is $r = 0$.)

cycle	mean gap m	std σ_*	A sim
234	-0.04996	0.03976	0.95135
546	-0.04597	0.03902	0.89056
858	-0.03117	0.03976	0.82842

11 Step 9 — Final SoH prediction and 95% band

$\text{SoH}_A^{\text{pred}} = \text{SoH}_A^{\text{sim}} + m$, band = $\text{SoH}_A^{\text{pred}} \pm 2\sigma_*$:

cycle	$\text{SoH}_A^{\text{pred}}$	-2σ	$+2\sigma$	real A exp
234	0.9014	0.8219	0.9809	0.88676
546	0.8446	0.7666	0.9226	0.82263
858	0.7972	0.7177	0.8768	0.79345

All three measured A points fall inside the 95% band, and the predicted curve sits close to the real cold-cell experiment — using only D and F data.

12 Step 10 — Log marginal likelihood

The log marginal likelihood [3, Eq. 2.30] is

$$\log p(y \mid X) = \underbrace{-\frac{1}{2} y^\top \tilde{K}^{-1} y}_{\text{data fit}} - \frac{1}{2} \underbrace{\log |\tilde{K}|}_{\text{complexity penalty}} - \underbrace{\frac{N}{2} \log 2\pi}_{\text{constant}}.$$

Numerically: $-0.402 + 19.278 - 5.514 = \boxed{+13.36}$.

Training maximises this over $\{\sigma_f, \ell_1, \ell_2, \sigma_n\}$, a bound-constrained problem (each hyperparameter must stay positive) solved here with a trust-region method [1], run from several restarts. The data-fit term rewards matching y ; the log-determinant term penalises over-flexible kernels — an automatic Occam’s razor requiring no separate validation set [3, Sec. 5.4].

13 Step 11 — Final score (RMSE)

$$\text{RMSE} = \sqrt{\frac{1}{M} \sum_i (\hat{y}_i - y_i)^2} \quad \text{against the real A experiment.}$$

Raw simulation (0.95135, 0.89056, 0.82842 vs 0.88676, 0.82263, 0.79345):

$$\text{RMSE} = \sqrt{\frac{1}{3}(0.06459^2 + 0.06793^2 + 0.03497^2)} = 0.0578.$$

GP prediction (0.9014, 0.8446, 0.7972):

$$\text{RMSE} = \sqrt{\frac{1}{3}(0.01463^2 + 0.02196^2 + 0.00380^2)} = 0.0154.$$

Final results (real data, 3-point subset)

RMSE raw simulation (no correction)	0.0578
RMSE GP-corrected prediction	0.0154
Measured points inside 95% band	100%
Log marginal likelihood	+13.36

The correction cuts the error from ≈ 0.058 down to 0.015 (about a $4\times$ improvement), and every point lands in the band. The full 16-point model reaches $\text{RMSE} \approx 0.008$.

The GP, being a smoother, pulls a little toward the average but still tracks the cold-cell experiment well. It is far better than the raw simulation, and it reports its own uncertainty so we know how much to trust each point.

14 I/P and O/P

Inputs: the D and F gaps (the only training data), plus Cell A’s simulation, cycle numbers, and temperature. **Output:** Cell A’s predicted experimental SoH curve with a 95% uncertainty band. The real measured A data entered only in Step 11, to grade the result — never to make it. So this is an honest held-out prediction, not a fit.

One sentence: learn the simulation’s error on two known cells, carry it to the cold cell along the temperature ($1/T$) direction with a Gaussian Process (which also reports its uncertainty), add it back to the cold cell’s simulation, and check against reality.

References

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